

Identifying constants on an open overlap

MathlibAnnex.aeConstants_eq_of_open_overlap

theorem

Uses positive measure to find a point where both almost-everywhere equalities hold.

Statement

Let X have a topology and a measurable structure, and let μ be a measure that assigns positive measure to every nonempty open set. Let Y be any set, $u : X \rightarrow Y$, and $c, d \in Y$. Suppose $V, W \subseteq X$ are open, $V \cap W \neq \emptyset$, and

$$u = c \quad \mu|_V\text{-almost everywhere}, \quad u = d \quad \mu|_W\text{-almost everywhere}.$$

Then $c = d$.

Assumptions

No topology or measurable structure on Y is required. There is no Haar, finite-measure, or Borel-compatibility assumption. Openness and nonempty overlap are used through positivity of that overlap, rather than through an assumed measurable-set identity.

Conclusion

The two constants coincide as elements of Y . A selected point of $V \cap W$ need not satisfy either equality; positivity permits choosing a point outside both exceptional sets.

As throughout the gluing argument, $u = c$ almost everywhere on V means almost everywhere for the restricted measure $\mu|_V$. The measure of the overlap may be infinite; only its being nonzero is needed.

Proof route

Restrict both equalities to the intersection, combine them there, and use positive measure to obtain a common good point.

▼ Proof steps

1. **Restrict the equalities.** Put $O = V \cap W$. Monotonicity of restriction transports the first almost-everywhere equality from V to O , and the second from W to O . Their conjunction holds almost everywhere for $\mu|_O$:

$$u(y) = c \text{ and } u(y) = d \quad \text{for } \mu|_O\text{-almost every } y.$$

Taking this finite conjunction is legitimate without any countable-cover assumption.

2. **Choose a good point.** The set O is nonempty and open, so $0 < \mu(O)$. The positive-measure existence lemma applied to the preceding conjunction supplies a point $y \in O$ at which both equalities hold. Therefore

$$c = u(y) = d.$$

This is the step that would fail if the overlap merely contained a point but had measure zero.

Main citations

- Exact declaration and its source — `MathlibAnnex.aeConstants_eq_of_open_overlap`
- Almost-everywhere equality on a restricted measure — `MathlibAnnex.AEConstantOn`

Lean source signature (exact)

```
theorem aeConstants_eq_of_open_overlap
  {u : X → Y} {V W : Set X} {c d : Y}
  (hVo : IsOpen V) (hWo : IsOpen W) (hVW : (V ∩ W).Nonempty)
  (hc : AEConstantOn μ u V c) (hd : AEConstantOn μ u W d) : c = d
```

In the source	Mathematical meaning
<code>[MeasurableSpace X] [TopologicalSpace X] (μ : Measure X) [μ.IsOpenPosMeasure]</code>	The ambient measure gives positive measure to every nonempty open set. These are surrounding source binders, including the positivity instance.
<code>{u : X → Y} {V W : Set X} {c d : Y}</code>	The function into an arbitrary value set Y , the two subsets, and constants c, d .
<code>(hVo : IsOpen V) (hWo : IsOpen W) (hVW : (V ∩ W).Nonempty)</code>	Both sets are open and their intersection is nonempty; consequently that open overlap has positive measure.
In the source	Mathematical meaning
<code>(hc : AEConstantOn μ u V c) (hd : AEConstantOn μ u W d)</code>	$u = c$ for $\mu _V$ -almost every point and $u = d$ for $\mu _W$ -almost every point.
<code>: c = d</code>	The output equality of the two values c, d . Positivity supplies a point outside both exceptional sets; an arbitrary overlap point is insufficient.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 12–15:

```
noncomputable section
open Set MeasureTheory Filter TopologicalSpace
```

```
open scoped ENNReal Topology
namespace MathlibAnnex
```

Exact source lines 49–52:

```
section Topological
variable {X Y : Type*} [MeasurableSpace X] [TopologicalSpace X]
variable (μ : Measure X)
```

Exact source lines 90–90:

```
variable [μ.IsOpenPosMeasure]
```

▼ Exact content identity

Declaration: MathlibAnnex.aeConstants_eq_of_open_overlap

Accepted content SHA-256: 4e5de888bc0c487988483f96f900779ad93931a4d8d21fc6add9be9d938246f3

Accepted source guide SHA-256: dfdbe70d1af7ebaba4806aeb68fd5e86d82bc7bac0050ab42fe7350de7477119

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73